

Allocation of internal cash flows- Evidence from an emerging market

Vishnu.K ¹, Aravind Sampath ²

¹ PhD student, Indian Institute of Management Kozhikode, India, E-mail: vishnuk10fpm@iimk.ac.in

² Assistant Professor, Indian Institute of Management, Kozhikode, India, E-mail: aravinds@iimk.ac.in

Abstract: How do firms in emerging markets allocate their most important source of funds, -internal cash flows? Our study analyzes this question in the context of an emerging market and our results indicate that on an average, firms' channel much of their contemporaneous internal cash flows through the capital structure in terms of repayment of borrowings. However, on a long-term basis, borrowings constitute a major source of funds for investments. Our study also finds evidence supporting that financially flexible firms (firms with less borrowings to asset ratio) invests less than the highly levered firms. Our results underscore the importance of borrowed funds as an important source of finance in developing countries where raising finance via equity issues is relatively costly.

1. Introduction: Hennessy and Whited (2007) state that, "corporate finance is primarily the study of financing frictions". Financial frictions are constraints that deter firms from making desirable investments. Examples are taxes, transactions costs, information asymmetry, bankruptcy costs and agency costs. In the absence of financing frictions, firms can costlessly access external capital markets and raise the required funds hence they don't need to save internal funds for maintaining financial slack (Keynes, 1936). Also, the empirical failure of standard model of investment with convex adjustment costs to predict the change in investment rates driven by changes in Tobin's q has been taken as a prima facie evidence of the presence of financial frictions at the firm level (Cao et al., 2013). The presence of financial frictions makes investment more sensitive to internal sources of finance which implies firms can actually tap the opportunities created by financial frictions by maintaining more liquid assets portfolio (Almeida et al., 2011). The market imperfections in terms of financial frictions, thus, creates a wedge between external and internal funding costs and this is why large fractions of projects are funded with internal funds (Brunnermeier et al., 2011). Internal funds generated has three major uses, namely, increasing investments, reducing external funds and pay-out decisions. These 3 alternatives, but with two degrees of freedom recognises the interdependence between financing and investment decisions which is largely ignored in the domain of empirical corporate finance¹. However, in recent times, studies recognize the interdependence and intertemporal nature of financing and investment decisions (e.g., Pulvino and Tarhan (2006), Gatchev et al. (2009), Gatchev et al. (2010), Dasgupta et al., (2011), Chang et al., (2014) Lewellen and Lewellen (2016)), and Drobetz et al., (2017)).

When firms embark upon financing and investment choices, the inevitable interaction between the choices impact decisions at the level of capital budgeting

¹ James Tobin on his discussion on the paper by Fazzari et al., (1988), stresses that the firm jointly determines investment, dividend payments, and other ways of allocating its cash flow.

rules, funds management policies, investment criteria and pay-out policies (Denis(2011), Almeida et al. (2011)). Thus, the presence of financial frictions can make internal funds more valuable and usage of internal funds is a joint decision about investment, financing and dividend decisions (Lambrecht & Myers(2013). In our article, we recognise this interconnectedness of internal funds (internal cash flows) by using it in a multivariate setting. This helps in reducing problem of omitted variables, thus, enhancing the precision of estimates as well as tracing the complete allocation of internal cash flows. Following (Dasgupta et al., 2011) and (Chang et al., 2014) , our empirical methodology thus imposes a restriction based on the accounting identity that sources of the cash should be equal to their applications. This identity-based system of equations acknowledges the interdependence of financial choices, thereby producing efficient coefficient estimates. Apart from this identity-based estimation, we use a unique setup where institutional voids increase the financial frictions further making internal funds more crucial.

In our article, we use data from an emerging economy, India, tapping into the unique institutional characteristics. First, unlike the developed economies like the U.S and U.K, the Indian capital market is underdeveloped, hence restricting the ability of firms to raise funds via equity issues due to transaction costs involved (Allen et al.(2012) and Komera et al.,(2018)). Also, the public debt market is at a rudimentary stage characterized mostly by illiquid assets(Gopalan et al. (2019)). Due to these structural limitations (institutional voids), Indian firms predominantly dependent on the private debt (bank borrowings) when it comes to borrowings (Allen et al. (2012) and G. S. Chauhan (2017)). However, this also comes with its idiosyncrasies, since the banking system is mostly dominated by public sector banks where access to credit is dependent upon various factors ((Srinivasan and Thampy(2017)). The lack of a mature capital market coupled with dependence on banking system results in a two-fold impact. First, this leads to higher transaction costs in raising funds via security issuance. Second, bank finance would be available only for firms that are large and established with good political connections (Beck et al., 2005). Thus, in contrast to developed economies where the scope of raising capital through external markets are broad, Indian firms are forced to rely on internal funds for various business activities (Gopalan et al.(2007). The prevalence of business groups is another unique feature which can act as an internal capital market that can help firms to a certain extent for their funding requirements(Lensink et al.,2003). In the Indian context, empirical evidence indicates that business groups facilitates transferring of funds to financially weaker firms(Gopalan et al., 2007). On the other hand, research also finds that group affiliated firms engage in tunnelling activities which results in minority shareholders' fund expropriation (Bertrand et al., 2002),(Kali & Sarkar, 2011).

This idiosyncratic institutional framework provides a good setting for analysing the allocation of internally generated cash flows in India. Unlike mature economies, the institutional setting of India itself restrains the firms from raising funds from capital markets. This, in turn, shapes the usage of funds in line with future business expectations and constraints. Given this unique institutional setup, we investigate the cash flow sensitivities to various decisions like cash holdings, investments, borrowings, share issues and dividend thereby tracking the complete usage of internal cash flows. Our investigation is also motivated by some empirical regularities that has been reported in the realm of Indian corporate finance.

Recent studies about Indian firms shows that investments are declining and due to this there is a trend of corporate deleveraging amongst firms (G. S. Chauhan, 2017). The reduction of investments has been mainly concentrated on firms with severe financial frictions (Das et al., 2017). It is also argued that the strengthening of creditor's rights due to passage of various laws has resulted in decline of financial leverage in Indian firms (Vig, 2013).

Taking together the above findings, it could mean that firms are reducing investments, dissaving and are primarily engaged in deleveraging. We try to verify these empirical findings by exploiting our methodological refinements since the fall in investments, low cash reserves and deleveraging could be driven by their joint sensitivities to internal cash flows. We analyse to what major uses (investments, deleveraging, increasing cash hoards & dividend) internal cash flows are allocated. We also analyse, how firm- characteristics affects the allocation of internal cash flows and finally is there a significant difference in allocation of internal cash flows between standalone firms and firms affiliated with business groups.

Our findings are as follows. First, firms allocate more than 50 % of current internal cash flows for the repayment of borrowings. Remaining funds are used for increasing CAPEX (16%), adding to cash reserves (12%) and for making other investments (11%). Share buy-backs and dividends account for 6 % of the internal cash flow allocations. On a long-term basis too, retirement of borrowings and increasing investments constitute the major use of internal cash flows by an average firm. Our findings also indicate that borrowings are the major source of finance for investment unlike the equity issues. Second, consistent with literature we find that CAPEX and dividends are relatively persistent while the mean reversion trends in cash holdings indicate that firms save more cash than required and gradually depletes it over a period. Third, firms which are highly levered (financially fewer flexible firms) makes more real investments in terms of CAPEX, than the firms that are less levered (financially flexible firms) which saves more cash out of internal cash flows. This result is surprising since recent empirical evidence from developed markets suggest that highly levered firms makes less investments and increases their cash buffer (Denis & Sibilkov, 2010). Fourth, though unconstrained firms (large size firms) and constrained firms (small size

firms) allocate about half of their current cash flows for repayment of borrowings, in long run unconstrained firms borrows more and invests more than the constrained firms. Finally, group affiliated firms make more investments than the stand-alone firms owing to the former's better access to debt market. Overall, our findings suggests debt cash flow sensitivity is higher than the investment cash flow sensitivity and faced with an increase(decrease) in current cash flow, firms decrease(increase) dependence on borrowings but later on increase the investments by the way of borrowings. Our results thus highlights, how borrowings react endogenously to the exogenous changes in internal cash flows.

2. A brief overview of related literature on the cash flow sensitivities to various decisions and the Indian setting.

Investment-cash flow sensitivity: In the Modigliani and Miller world(1958), capital structure is irrelevant to the value of a firm and in this case there is a perfect substitutability between internal funds and external funds. Due to the market imperfections in the form of taxes, issuance costs, agency cost, bankruptcy costs, there exists a wedge between the cost of internal and external funds and internal sources of funds becomes cheaper. For instance, due to the information asymmetry about asset values and investment opportunities firms follow a pecking order in the source of funds in which firms fund their positive NPV projects with internal financing first and then go for less sensitive security-debt followed by all kinds of hybrid debt with equity components and finally with external equity as a last resort as argued by (Myers & Majluf, 1984). Another way in which issuing debt securities is better than equity issues is that former helps in reducing agency costs(Jensen,1976). On the other hand, the asymmetric information about the riskiness of assets lead to increased debt issuance costs and such firms end up issuing more equity than debt (Stiglitz and Weiss (1981)), De Meza and Webb (1987). Thus, informational asymmetries, costly monitoring, contract enforcement, and incentive problems lead to an imperfect substitutability between internal and external funds and due to this internal fund are cheaper than external funds. Internal funds can be used for investment purposes apart from other uses like reducing external finance thereby sparing debt capacity as well as paying dividends. Of this literature has extensively focused on how sensitive investment to internal cash flows is. Pioneered by (Fazzari et al., 1988), the literature on investment cash flow sensitivity has analysed how sensitiveness of investment to cash flows can be taken as an evidence for financial constraints. Later, Kaplan & Zingales(1997) rejects the view that investment cash flow sensitivity is an indication of financial constraints since they find that those firms that appear less financially constrained exhibit significantly greater sensitivities than firms that appear more financially constrained. They redefined the question that, given the fact that investment is sensitive to cash flows empirically, what are the drivers of investment-cash flow sensitivity (Kaplan and

Zingales ,2000, p. 711). Investments are related to cash flows mainly because of two reasons, there exists a correlation between cash flows and investment opportunities and due to cheaper costs of internal funds, opportunistic managers may overspend internally available funds (Lewellen & Lewellen,(2016). Recently it has been established that investment cash flow sensitivity has declined to a great extent in developed markets due to growing wealth of economies worldwide and relaxation of financial constraints at the firm level, though in developing markets, investment-cash flow sensitivity is still prevalent (Larkin et al., 2018).

Debt cash flow sensitivity: Internal cash flows can be also used to retire debt and thereby sparing the debt capacity. As conjectured by (Hovakimian et al., 2001) firms may use past cash flows to pay down debt leaving them temporarily under levered. This was supported by (Byoun, 2008), who argues that firms with positive cash flow shocks are more likely to retire debt than equity if the adverse selection costs of the latter is higher than the former. Faulkender et al., (2012) finds that firms with large (positive or negative) operating cash flow make more aggressive changes in their capital ratios than firms with similar leverage deviations but cash flow realizations near zero. Similarly, Daniel et al., (2008) finds that firms that register positive cash flows uses about $\frac{1}{4}$ th of their cash flows to reduce external financing by retiring debt and/or repurchase equity.

Equity cash flow sensitivity: Firms also uses their cash flows for share buy-backs or stock repurchases. In theory, if excess cash flows are temporary or unstable, firms should go for repurchases than dividends as dividends tend to be sticky in nature. Also as pointed by Brockman & Chung (2001), “Firms with relatively large cash flows are more likely to repurchase shares for the purpose of wealth distribution and thereby reducing agency costs”. Recently share repurchases has considerably grown over dividends(Koijen & Van Nieuwerburgh, 2011) & established corporations distribute more of their cash flows through repurchases and less through dividends(Grullon & Michaely, 2002)

Dividend cash flow sensitivity: As per Miller & Modigliani,(1961), in the presence of perfect markets, dividend policy is irrelevant to firm’s value. However, in practice, corporations pay dividends and irrelevance theory has been rejected empirically (DeAngelo & DeAngelo, 2006). Corporations pay dividends from operating cash flows and one explanation of this is based on signalling theory which predict that dividend changes convey information about current or future cash flows (Bhattacharya, 1979). Another explanation is based on over-investment hypothesis of (Jensen, 1986) and (Easterbrook, 1984) that firms pay dividend to reduce free cash flow problems which may be wastefully invested by opportunists managers. However due to sticky nature of dividends, firms rarely cut the dividends (Daniel et al., 2008) even when faced with a cash short falls and in such instances firms go for security issues to finance pay-outs(Farre-Mensa et al., 2014).

Cash flow sensitivity of cash: Cash flow sensitivity of cash research was pioneered by (Almeida et al., 2004). They find that firms that are financially

constrained exhibits higher propensity to save cash from cash flows. In similar lines, for an international sample of firms, (Khurana et al., 2006) finds that sensitivity of cash holdings to cash flows decreases with financial development. However, (Riddick & Whited, 2009) find that savings and cash flow are negatively related because firms lower cash reserves to invest after receiving positive cash-flow shock. Bao et al., (2012) argues that both cash savings and cash dis-savings can happen based on asymmetric cash flow sensitivity.

2.1 Interconnectedness of corporate decisions: Though the above decisions reflect cash flow sensitivity for various financial and investment decisions, these were looked in isolation or as standalone decisions without acknowledging the fact that above decisions are interdependent. For example, a firm may use less internal resources for its investment purposes and as per conventional ICFS literature, the lack of investment cash flow sensitivity indicates absence of financial constraints. However the same firm may be saving more cash from cash flows (precautionary cash savings) and this is an indication of financial constraints as per (Almeida et al., 2004). Though the interdependence of financial decision was pointed by Tobin (1988) in his discussion of the paper by (Fazzari et al., 1988), the literature until recently, has neglected this interconnectedness. To the best of our knowledge, it is Pulvino & Tarhan, (2006), who examines corporate decisions simultaneously, subject to the constraint that sources of cash equals use of cash. They argued that reduced form of equations suffers from omitted variable bias and lacks economic interpretation. Based on their simultaneous equation set up, they find that investment is less sensitive to cash flows and are insignificant whereas financing-cashflow sensitivities are large and highly significant, regardless of the firm's financial health. In similar lines, (Gatchev et al., 2009), (Gatchev et al., 2010), (Dasgupta et al., 2011) analyse the investment and financial decisions in a joint setting and their findings suggests that firms use much of their contemporaneous cash flows for deleveraging purposes thereby sparing the debt capacity. (Chang et al., 2014) decomposes cashflow into transitory (random part) and permanent components (deterministic part) and they find that constrained firms save more from transitory cash flow to build their cash reserves and allocates less for investment purposes. Similarly, (Lewellen & Lewellen, 2016) bifurcates cash flows into expected and unexpected part and they find that unconstrained firms exhibit higher investment-cash flow sensitivity and is largely driven by their reactions to unexpected cash flow. (Drobetz et al., 2017) using multi-equation model of cash flow sensitivities analyse how liquidity crisis affects corporate decisions. They find that recent financial crisis affected financially weak firms the most, leading them to cut down their investment decisions also forcing firms to trim their balance sheet.

2.2 Why internal cash flows are important for a country like India? The role of internal source of funds like cash flows in a country like India cannot be overstated given the market imperfections in terms of poorly functioning institutions, leading to severe agency and information problems (Khanna &

Palepu, (2000). The importance of internal sources of finance for developing economies like India is reasoned by the logic that *“fragmented capital markets retard the efficient allocation of resources and force business firms to rely on internal sources of funds. Also capital and capital goods market in less developing countries are imperfect and financial infrastructure is inadequate”* (Athey & Laumas, 1994). Similarly, (Allen et al., 2012) *finds that internal finance accounts for 45% of total (annual) financing, while alternative finance accounts (finance from non-banking and banking sources) for 30% of financing.* Bank loans dominate the credit market in India due to lack of sound debt market (G. S. Chauhan, 2017). However, in India, bank based finance is available only for large firms which has strong relationships with banks (Banerjee & Duflo, 2014) (Srinivasan & Thampy, 2017). The importance of Internal source of funds is clear from the fact that India is devoid of developed debt market (Gopalan et al., 2019) (V. S. Khanna & Varottil, 2012) and equity market is accessible to large firms even though large Indian firms rely less on equity financing (Allen et al., 2012). For smaller and young firms, the access to credit is even tighter and hence internal source of funds is the most prominent source. Thus, nascency of public debt market, and over dependence on commercial banks for credit purpose suggests that, in general firms in India are a credit constrained and these highlights the role of internal cash flows which could affect the progress of the economy.

2.3 Role of BG/Internal Capital Markets (ICMS) in alleviating financial constraints: As mentioned above owing to weak institutional mechanism and resulting market failures to supply adequate credit to firms in India, Firms affiliated with business groups can benefit from access to these internal institutions to mitigate external market failures (Khanna & Palepu, 2000), (Desai et al., 2004). Apart from the financial intermediation between parents and subsidiaries, the former can also discipline the latter in the proper usage of funds. On the other hand, ICMS are also alleged for expropriating minority shareholders by tunnelling resources from firms where they have low cash flow rights to firms where they have high cash flow right there by creating inefficiencies (Bertrand et al., 2002).

3. Methodology: Traditional models in finance view decisions in isolation. Extant literature on investment cash flow sensitivity (ICFs), usually employs the reduced-form investment equations where the dependent variable represents investments in fixed assets for the firm i during the period t with the variable of interest (cash flows) and other control variables as explanatory ones². Similarly, Frank and Goyal(2003) use an empirical specification which involves a single financing choice (long term debt issues) as a dependent variable with a single variable of interest (net financing deficit) along with the controls. The problems with such static-single equation models is that they fail to acknowledge the identity where sources of funds equal uses of funds. Hence the estimations may be inefficient leading to biased coefficients(Gatchev et al., 2010). We overcome this methodological flaw of ignoring the accounting identity, where sources of funds is equal to its uses by following (Dasgupta et al., 2011) to analyse the uses of internal cash flows³. Cash flows can be used to invest in fixed assets, increase cash reserves, to retire external finance (debt and equity) and for the pay-out decisions (dividends and share repurchases) thereby tracking the complete allocation of the internal cash reserves. We try to track how internally generated one rupee (rupee or INR is the official Indian currency) is allocated to different uses.

We use the basic cash flow identity to track the sources and uses of cash.

$$CFO_t + CFI_t + CFF_t = \Delta CASH_t \quad (1)$$

Where,

CFO_t =Net cash flow from operations at the time “t”.

CFI_t = Net cash flow from investment activities at the time “t”.

CFF_t =Net cash flow from financing activities at the time “t”.

Note that net implies, cash inflows and cash outflows adjusted. Thus, a positive CFI_t (net cash flow from investment activities) means firms had a positive cash inflow from investment activities, implying firms had more asset sales(cash comes in) than asset purchases(cash goes out). Similarly, a positive CFF_t (net cash flow from financing activities) implies firms raised more funds (cash comes in) via security issues than the payments made by firms (cash outflows). Hence delta cash reflects the effect of the combined cash inflow or outflow from the above three activities.

² See for e.g. Fazzari et.al ,(1988), Kaplan & Zingales (1997), Ağca & Mozumdar (2008), Bushman et al., (2011), Chen & Chen (2012), & Moshirian et al., (2017)

³ The difference between the methodology of (Gatchev et al., 2010) & (Dasgupta et al., 2011) is that in former, cash flow identity do not hold in data, hence constraints are externally imposed while in latter, data is in the form of cash flow identity, hence external constraints are redundant. Nevertheless, both methods assume that internal cash flows are exogenous where are as the uses of cash flows (investment, retirement of external finance and payment of dividends) are endogenous. This is a departure from earlier models in finance which assumes net financing deficit as exogenous one, for example, Frank and Goyal (2003), & Gatchev et al.,(2009)

Equation (1) is split into various components that constitute operating, investment and financing decisions.

CFO_t : This is the net income adjusted with non-cash items and non-operating items. **CFO_t** denotes cash flows which in our context implies the internally generated source of funds. Unlike previous studies which use income before extraordinary items (“profits”) plus depreciation, we follow (Bushman et al., 2011) to define cash flows(**CF_t**)⁴. Our definition of cash flows (**CF_t**) thus excludes non-cash net working capital (NWC).

CFI_t can be split into two

1. Net cash inflow or outflow from investments in fixed assets and capital work in progress, our proxy for real investments (**CAPEX**).
2. Net cash inflow or outflow made to acquire shares, warrants or debt instruments of other enterprises and interests in joint ventures and net cash inflow or outflow related to other investments in terms of loans given (repaid) to (by) subsidiaries, cash set as restricted cash etc. (**OTHER INVESTMENTS**).

CFF_t can be split into three

1. Cash inflow or outflow (repurchases, share buybacks) from share issues, hence net issues (**EQUITY**).
2. Cash inflow and outflow from borrowings (adjusted with interest payments made) (**BORROWINGS**)
3. Dividends paid (including dividends tax) (**DIVIDENDS**)

Hence equation (1) can be re-written as

$$CFO_t + CAPEX_t + OTHER\ INVESTMENTS_t + EQUITY_t + BORROWINGS - DIVIDENDS_t = \Delta CASH_t \quad (2)$$

⁴ Refer Bushman et al., 2011 to know how accrual component in Cash flow from operations can influence the results.

Rearranging 2 gives

$$CFO_t = \Delta CASH_t - CAPEX_t - OTHER\ INVESTMENTS_t - EQUITY_t - BORROWINGS_t + DIVIDENDS_t \quad (3)$$

Now, based on the equation (3), the interpretation is that one rupee generated as cash flow from operations must be equal to its various uses like investments and financing decisions. Negative sign of asset items (CAPEX & OTHER INVESTMENTS) indicate cash inflows from asset sales. Similarly, negative signs for financing sources (Equity and Debt) denotes cash inflows from security issues. Finally, positive sign for dividends indicates cash outflow⁵. For the ease of interpretation, we write equation (3) as

$$CFO_t = \Delta CASH_t + CAPEX_t + OTHER\ INVESTMENTS_t + EQUITY_t + BORROWINGS_t + DIVIDENDS_t \quad (4)$$

Equation (4) implies that one rupee of internal cash flows can be used to increase the cash reserves, increase investments in CAPEX and other investments, retire debt and equity and increase pay-outs. Thus, funds can be used either to increase the assets base, decrease the liabilities or pay out as dividends⁶.

From equation (4) we have the following standalone equations.

$$\Delta CASH_{it} = \alpha_{it} + \beta_1 CFO_{it} + \gamma CONTROLS_{it} + \varepsilon_{it} \quad (5)$$

$$CAPEX_t = \alpha_{it} + \beta_2 CFO_{it} + \gamma CONTROLS_{it} + \varepsilon_{it} \quad (6)$$

$$OTHER\ INVESTMENTS = \alpha_{it} + \beta_3 CFO_{it} + \gamma CONTROLS_{it} + \varepsilon_{it} \quad (7)$$

$$BORROWINGS_{it} = \alpha_{it} + \beta_4 CFO_{it} + \gamma CONTROLS_{it} + \varepsilon_{it} \quad (8)$$

$$EQUITY_{it} = \alpha_{it} + \beta_5 CFO_{it} + \gamma CONTROLS_{it} + \varepsilon_{it} \quad (9)$$

$$DIV_{it} = \alpha_{it} + \beta_6 CFO_{it} + \gamma CONTROLS_{it} + \varepsilon_{it} \quad (10)$$

⁵ Dividends related to financing activities are only separately considered here. Dividends related to investment activities is adjusted with other investments. Likewise, interest received on loans made is adjusted with other investments whereas interest paid on borrowings is adjusted with debt

⁶ Positive signs of CAPEX, other investments denote increase in assets. This is because addition to assets denotes cash outflows denoted by negative sign in cash flow statements and if we write the equation in terms of CFO (see equation 4), sign reverses, implying positive sign for addition to assets. Similarly, positive sign for debt and equity denotes reduction of external finance as in cash flow statements same would be represented by negative signs.

Equations from 5 to 10 denotes the sensitivities of cash flows to various uses. For instance equation (5) is similar to the cash flow sensitivity of cash as proposed by Almeida et al.,(2004) and equation (6) is the investment cash flow sensitivity equation. By the virtue of the accounting identity that cash flow sources equal its uses, sum of all the cash flow sensitivities of the above standalone equations should add up to one. Similarly, the controls should add up to zero since any explanatory variable that increases or decreases the allocation of cash flows to one use will be offset by the allocation made to other use or uses because of the accounting identity constraints⁷. Thus

$$\sum_1^6 \beta = 1 \text{ \& } \sum \text{CONTROLS}_i = 0 \quad (11)$$

Dependent Variables

The dependent variables are endogenous in nature by construction (see equation (4)). Delta cash, CAPEX, other investments, debt, equity, and dividends (all are sourced from cash flow statements with adjustments) are the dependent variables which proxies for different uses of cash flows.

Independent Variables

Cash flow from operations is the main exogenous variable which is directly available from the statement of cash flows. Alternatively one can derive cash flows as profit after tax (PAT) adjusted for the effect of non-cash components like depreciation, accruals etc. (Bushman et al., 2011).

Control Variables.

Firm Size: We use natural log of total assets as a proxy for firm size. Larger firms tend to have economies of scale in operation, better transparency hence lower information asymmetry and lower asset volatility which results in more debt and equity issues (Byoun, (2008). Unlike the large firms, small firms are subject to information asymmetry, volatile cash flows hence restricted access to external capital market therefore more dependable on internal cash flows(Byoun,2007).

Leverage: Leverage is measured as total long-term borrowings to total assets. Higher the leverage, more chance for debt overhang which prevent firms to raise finance even to fund positive NPV projects (Myers, 1977). Also, Whited(1992)shows that investment is more sensitive to cash flow in firms with high leverage than in firms with low leverage. Also, leverage increases refinancing risk and firms hold cash to mitigate its adverse effects(Harford et al., 2014). Finally, firms with larger cash flows are more prone to

⁷ If MB, proxied for growth opportunities is a control variable and firms with higher MB allocates more cash flows to increase investments, less amount will be available for other uses which offsets the combined effect of MB to total uses as 0.

agency issues and in such cases higher leverage would force managers to pay out more rather than making overinvestments(Jensen, 1986).

MB ratio: MB ratio is widely used as a proxy for growth opportunities. It is also interpreted as a measure of superior managerial performance hence strong cash flows ((Lang et al.,(1989),(Servaes, (1991)). Higher Q ratios are associated with decision to issue equity(Huang & Ritter,2004) as well .We measure q ratio as $\frac{\text{Market value of equity} - \text{book value of equity}}{\text{book value of total assets}}$ /book value of total assets.

Sales Growth: We also control sales growth as it also controls for investment opportunities ((Cleary et al.,(2007), Faccio et al.,(2016)).

Profitability: Profitable companies face less constraints to raise funds from markets and hence we control for operating profitability measured as EBIT divided by net sales.

4. Data and Sample

We source our data from Prowess IQ database & Prowess dx database (vintage March 2020), maintained by the Centre for Monitoring the Indian Economy (CMIE). The prowess database is analogous to the Compustat database and provides firm-level data mainly compiled from annual reports of the companies (both listed and unlisted). Prowess Database has been used extensively by several studies focusing on Indian firms(see Khanna and Palepu, (2000); Gopalan, Nanda, & Seru (2007); Vig (2013) and Baghai and Becker, (2018)). Our sample covers firms listed and permitted companies in the major stock exchanges in India, during the period 2000-2019. We restrict our sample by excluding all financial firms (NIC codes 641-663), as well as state-owned firms (wholly or partially). We remove:

- a. Firm years for which total assets were negative, missing or 0.
- b. Firm years which has MB ratio greater than 10 to reduce measurement error following.
- c. Firm years with assets growth greater than 100 % to avoid large M&A transactions or bulk sale of equities.
- d. Firm years for which net worth is negative or missing or 0.
- e. Firm years for which sales growth is greater than 100 %.
- f. Firm years for which total assets is less than Rs 5 million to avoid small denominator problem.

Table 1

Descriptive statistics

Variable	n	Mean	S.D.	Min	0.25	Median	0.75	Max
Net cash flow from operation	19157	0.09	0.13	-0.33	0.01	0.08	0.15	0.53
CAPEX	19365	0.08	0.13	-0.1	0.01	0.04	0.1	0.7
Other Investments	19365	0	0.06	-0.21	-0.01	0	0	0.31
Net share issues	19365	-0.01	0.05	-0.36	0	0	0	0.04
Borrowings	19365	-0.01	0.13	-0.59	-0.03	0	0.06	0.31
Dividends	19365	0.01	0.02	0	0	0	0.02	0.14
Delta Cash	19365	0.01	0.06	-0.21	-0.01	0	0.01	0.27
MB ratio	21265	1.12	0.63	0.25	0.78	0.94	1.22	3.99
Sales growth	21265	0.09	0.16	-0.44	0.01	0.1	0.19	0.5
Log (total assets)	21265	7.66	1.91	3.76	6.28	7.58	8.92	12.69
Leverage	19024	0.3	0.17	0	0.17	0.3	0.43	0.71
Operating Profitability	21260	0.09	0.18	-1.11	0.04	0.09	0.14	0.66
Log(age)	29804	2.71	0.67	0.69	2.4	2.89	3.18	3.64

The data is from Prowess for listed companies during the period 2000–2019. The dependent variables are Delta cash, CAPEX, Other investments, Net share issues, Borrowings and Dividends. The main variable, net cash flow from operations (CFO) is directly taken from statement of cash flows of firms. CAPEX is defined as net cash flow (cash outflow adjusted with cash inflow) from investments in fixed assets and capital work in progress. Other investments denote net investments (inflows adjusted with outflows) made in shares and warrants or debt instruments of other companies in joint ventures as well loans given(taken) from subsidiaries. Net share issues are measured as net cash flow (inflows adjusted with outflows) from share issues. Borrowings denotes net cash flow (inflows adjusted with outflows) from borrowings (adjusted with interest payments made). Dividends paid is measured as cash outflow due to dividends paid including taxes. For the controls, we use a rolling 3-year average. MB ratio is defined as, [Market value of equity less Book value of equity add Book value of total assets]/Book value of total assets. Sales growth is the change in net sales scaled by lagged net sales. Natural log of the total book value of assets is the proxy for size. Leverage is defined as total debt (the sum of short-term and long-term debt) divided by total assets. Operating Profitability is EBIT divided by net sales. Log(age) is the natural log of age. We measure age based on the listing date on the stock exchange. All the variables, except, MB ratio, Sales growth, Operating profitability, Log(assets) & Log(age) are scaled by total assets. All the variables are winsorized at both tails (1% & 99%). This has led to the violation of cash flow identity

4.1 Descriptive statistics

The average firm reports a cash flow of 9%. Net share issues and Borrowings together contribute 2% of funds. Standard deviation of debt issues is greater than equity issues

which implies firms periodically borrows money and they do it large amounts. Major use of cash flow is investments in CAPEX. Remaining funds are allocated to increase cash reserves and to pay out as dividends ($1\% + 1\% = 2\%$). Other investments which comprises of long-term investments in securities and warrants of other companies, loans provided etc. has a mean zero indicating that firms invest (grant loans) and disinvests (get repayments) on an alternate basis. Except dividends, all other cash flow variables have a higher range.

4.2 Correlation analysis

Table 2 exhibits correlation coefficient between variables used in the study. All the dependent variables in the standalone equation exhibits positive association with the cash flow from operations. This implies that an extra rupee of cash flow results in adding to cash reserves, making investments in CAPEX and other investments, reducing external finance, and paying out as dividends. Size has a positive relationship with CFO which shows that firms with larger assets base can have higher cash flows. The negative association between CFO and leverage indicates that as cash flows increases, firms rely less on borrowings and vice versa. Our proxy variables for growth opportunities, MB ratio and sales growth have strong positive correlation between them as well as with cash flows which suggests that profitable investment opportunities embedded in cash flows is reflected in growth measures.

Table 2

Pearson correlation coefficients table -

	Net cash flow from operation	CAPEX	Other Investments	Net share issues	Borrowings	Dividends	Delta Cash	MB ratio	Sales growth	Log (total assets)	Leverage	Operating Profitability	Log(age)
Net cash flow from operation	1												
CAPEX	0.323***	1											
Other Investments	0.232***	-0.0139*	1										
Net share issues	0.0779***	-0.232***	-0.172***	1									
Borrowings	0.303***	-0.592***	-0.114***	0.0417***	1								
Dividends	0.321***	0.193***	0.0560***	-0.00593	-0.101***	1							
Delta Cash	0.223***	-0.00605	-0.123***	-0.143***	-0.0901***	0.0632***	1						
MB ratio	0.179***	0.0714***	0.0858***	-0.0254***	-0.00884	0.374***	0.0277***	1					
Sales growth	0.249***	0.360***	0.0732***	-0.144***	-0.182***	0.228***	0.0970***	0.104***	1				
Log (total assets)	0.107***	0.0928***	0.0690***	-0.0215***	-0.0332***	0.135***	-0.00590	0.218***	0.0705***	1			
Leverage	-0.0304***	0.105***	-0.0467***	-0.0508***	-0.0194**	-0.306***	-0.0467***	-0.176***	0.0477***	0.188***	1		
Operating Profitability	0.214***	0.174***	0.0514***	-0.0131*	-0.0469***	0.257***	0.0431***	0.162***	0.287***	0.216***	-0.0413***	1	
Log(age)	0.0165**	-0.0624***	-0.0138*	0.0628***	0.0511***	0.0451***	-0.00247	0.0997***	-0.0685***	0.0951***	-0.0563***	0.0275***	1

5. Main empirical analysis

Following (Dasgupta et al., 2011), we estimate equations 5 -10 in panel regression framework by employing firm and time fixed effects. Beta-coefficients of cash flows indicates the cash flow sensitivity of average firms to six uses. Table 3 exhibits regression results for the sample firms during the year 2000-2019. Column 1 to 6 indicates the cash flow sensitivities to various decisions. Column 7 reports sum of coefficients of cash flows and control variables.⁸ Out of 92 Rs generated, approximately 50 rupees is going for debt repayment. Thus, debt retirement constitutes the major use of the contemporaneous internal cash. The higher sensitivity of borrowings to internal cash flows also suggest that borrowings is a major source of funds for cash flow short falls. This is in line with findings reported by Daniel et al.,(2008) in the context of developed countries. Their study finds that debt is a major source of fund for managing cash flow shortfalls. Second major use of funds is investments in CAPEX projects (.16/.93). This coefficient is consistent with the current studies which report that firms in poor countries(based on GDP) reports average cash flow coefficients on investments which are all positive and hovers around 10%-20% (Larkin et al.,(2018)). Faced with a positive cash flows, average firm adds to cash reserves (.11/.93) and makes other investments (.11/.93). Thus about 40% ($0.112+0.156+0.105/.93$) of the funds are only used to increase the assets base. Remaining 60 % goes out of the business in the way repayment of borrowings, equity, and dividend pay-outs. The sensitivity of share issues to operating cash flows are less, and this indicates equity is not a major source of funds in case of cash flow short falls. This is in stark contrast with the developed economies where equity issues are majorly driven by cash flow shortfalls (Huang & Ritter, (2015). Only 1% of the funds are used for making dividend payments and the weak sensitivity of dividends to cash flows are similar to findings in developed countries (Gatchev et al., 2009). Regarding controls, MB ratio has a negative association delta cash which implies in the presence of growth opportunities firms use their cash reserves, raise funds by security issues and use the funds mobilized to increase investments. The negative relationship between leverage and delta cash could imply that the former reduces the agency costs associated cash holdings in countries like India where the governance mechanism is relatively weak (Y. Chauhan et al., 2018) and the negative association of leverage with investments suggests that debt overhang reduces investments on an average. Increase in leverage results in repayments of borrowings (coefficient is positive and statistically significant) to avoid the risks of bankruptcy but equities are issued to raise the funds. As firms become more profitable, they raise external finance by issuing equities and increase their borrowings, draw down their cash balances and use the proceeds to invest. When firms mature, they cut down their

⁸ Post winsorization, cash flow identity is violated and hence sum of beta coefficients on cash flow sensitivities is less than the required sum, 1. Similarly sum of the coefficients on controls is less than the required sum 0. Such violations are in line with existing studies(Refer Drobetz et al., (2017), Drobetz et al., (2019)

investments and pay dividends. In a nutshell, firms allocate much of their current operating cash flows to make changes in financing structure than the asset structure.

Table 3

Baseline regression results

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Debt issues (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.112***	0.156***	0.105***	0.039***	0.499***	0.016***	0.927
Robust Standard Errors	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.00)	
MB ratio (3-Year Average)	-0.002	0.020***	0.007***	-0.007***	-0.026***	0.006***	-0.002
Robust Standard Errors	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	
Sales Growth (3-Year Average)	0.013***	0.131***	0.003	-0.038***	-0.125***	0.012***	-0.004
Robust Standard Errors	(0.00)	(0.01)	(0.01)	(0.00)	(0.01)	(0.00)	
Log (total assets) (3-Year Average)	-0.011***	-0.038***	-0.005***	0.012***	0.044***	-0.003***	-0.001
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Leverage (3-Year Average)	-0.013**	-0.084***	-0.018***	-0.016***	0.160***	-0.029***	0
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Operating Profitability (3-Year Average)	-0.007*	0.071***	-0.004	-0.011**	-0.039***	0.010***	0.02
Robust Standard Errors	(0.00)	(0.01)	(0.01)	(0.00)	(0.01)	(0.00)	
Log(age) (3-Year Average)	0.003	-0.046***	0.009	0.01	0.006	0.009***	-0.009
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
constant	0.072***	0.481***	0.007	-0.128***	-0.417***	0.01	0.025
Robust Standard Errors	(0.02)	(0.06)	(0.02)	(0.02)	(0.06)	(0.01)	
R-square	0.059	0.176	0.049	0.046	0.24	0.156	
Number of observations	16088	16088	16088	16088	16088	16088	

The data is from Prowess for listed companies during the period 2000–2019. The dependent variables are Delta cash, CAPEX, Other investments, Net share issues, Borrowings and Dividends. The main variable, net cash flow from operations (CFO) is directly taken from statement of cash flows of firms. CAPEX is defined as net cash flow (cash outflow adjusted with cash inflow) from investments in fixed assets and capital work in progress. Other investments denote net investments (inflows adjusted with outflows) made in shares and warrants or debt instruments of other companies in joint ventures as well loans given(taken) from subsidiaries. Net share issues are measured as net cash flow (inflows adjusted with outflows) from share issues. Borrowings denotes net cash flow (inflows adjusted with outflows) from borrowings (adjusted with interest payments made). Dividends paid is measured as cash outflow due to dividends paid including taxes. For the controls, we use a rolling 3-year average. MB ratio is defined as, [Market value of equity less Book value of equity add Book value of total assets]/Book value of total assets. Sales growth is the change in net sales scaled by lagged net sales. Natural log of the total book value of assets is the proxy for size. Leverage is defined as total debt (the sum of short-term and long-term debt) divided by total assets. Operating Profitability is EBIT divided by net sales. Log(age) is the natural log of age. We measure age based on the listing date on the stock exchange. All the variables, except, MB ratio, Sales growth, Operating profitability, Log(assets) & Log(age) are scaled by total assets. All the variables are winsorized at both tails (1% & 99%). This has led to the violation of cash flow identity. The regression equations are estimated with firm and year fixed effects. To save the space year dummies are not reported. Robust standard errors are presented in parentheses. Coefficients significant at the 10%, 5%, and 1% levels are indicated by *, **, and ***, respectively. All equations are estimated using the single-equation model.

5.1 Inter-temporal nature of financing and investment decisions: Our baseline results are based on the interdependence of financing decisions based on the cash flow identity. In this section we analyse the inter-temporal nature of financing and investment decisions. Recognition of intertemporal dependence between corporate financing and investment decisions is important since future period financing and investment decisions are impacted by today's decisions which can provide new insights (Denis, 2011). We try to incorporate the intertemporal dependence between cash flows and financing and investment decisions by adding 2 leads of cash flows to our setting based on the theoretical and empirical framework suggested by (Dasgupta et al., 2011) and (Lewellen & Lewellen, 2016). We also extend our model by adding lagged values of our dependent variables to capture the persistence/inertia of the cash flow variables.

Table -4 presents the intertemporal treatment of cash flow allocations. Like our base line results, current cash flows are highly sensitive to debt issues followed by CAPEX, other investments, and cash. To account for the long-term cash flow allocations, we have added 2 lagged values of cash flows. The results indicate that firms continue to make investments in the subsequent periods too. Firms allocate Rs .139/.938 of current year cash flows for CAPEX investments and in the next 2 years ($t+1$ & $t+2$) firms spend Rs .101(.052+.049) which makes the cumulative amount spend on investments as .24(.139+.052+.049). Major source of funds for meeting the investments comes from debt issues (Rs .084+.039) and firm save a part of the funds raised to increase their cash reserves also (Rs .038). In the year $t+2$, firms spend Rs .049 for investments, pay Rs .008 as dividends and invests Rs .006 in other assets. To meet the funds, firms draw down on their cash balances, issue debts and equity. Thus, over a period of three years, firms invest Rs .13 in cash reserves (.108+.038-0.016), Rs .24 in CAPEX (.139+.052+.049), & Rs .134 in other investments. Hence out of Rs .985 generated as internal cash flows during the 3-year period, Rs .504 is used for investments in assets and balance Rs.481 is used for retiring debt, equity, and payment for dividends. Retirement of debt and increasing investments constitute the major use of internal cash flows by an average firm on a long-term basis. To account for persistence, we also added lagged dependent variables as our independent variables. The variables that exhibit relatively higher persistence in the results are CAPEX (.218) & dividends (.216). Due to sticky nature of dividends, persistence of dividend is unsurprising. The persistence of CAPEX is in confirmation with our results which indicates that firms keeps increasing investments over a period. The reversion trends in cash holdings indicate that firms save more cash than required and gradually depletes it over a period. 1 Rs saved today will lead to using .29 Rs tomorrow. Saved cash is used mainly for increasing investments Rs .161(.107+.054) and for the repayment of borrowings. Other investments are mean reverting in nature, and this is because other investments include inter-group loans which involves periodic repayments. Unlike equity issues, borrowings show some degree of persistence. 1 Rs borrowed today will lead to Rs .04 borrowings tomorrow. The lack of mean reversion in borrowings may be due to lack of transactions costs involved in borrowings compared to

equity issues. Since firms borrow mainly from banks which is of private nature, less transaction costs will be there unlike equity issues which involves floatation costs⁹.

⁹The reversal in net share issues is consistent with the notion that firms incur fixed costs when they tap equity market and hence firms tends to issue more equity than needed for funding current projects.

Table 4

Intertemporal effects on cash flow sensitivities

Intertemporal effects on cash flow sensitivities

[illegible]

Sales Growth (3-Year Average)	0.023***	0.083***	0.009	-0.035***	-0.093***	0.009***	-0.004
Robust Standard Errors	(0.00)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)	
Log (total assets) (3-Year Average)	-0.013***	-0.039***	-0.002	0.010***	0.050***	-0.002***	0.004
Robust Standard Errors	(0.00)	(0.00)	(0.00)	(0.00)	(0.01)	(0.00)	
Leverage (3-Year Average)	-0.020**	-0.099***	-0.020***	-0.015**	0.173***	-0.022***	-0.003
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Operating Profitability (3-Year Average)	0.006	0.033***	-0.009	-0.004	-0.017	0.007***	0.016
Robust Standard Errors	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)	
Log(age) (3-Year Average)	0.005	-0.017	0.006	0.002	-0.014	0.009***	-0.009
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.01)	(0.00)	
constant	0.097***	0.432***	-0.016	-0.084***	-0.431***	0.005	0.003
Robust Standard Errors	(0.03)	(0.05)	(0.03)	(0.03)	(0.06)	(0.01)	
R-square	0.148	0.274	0.064	0.041	0.305	0.22	
Number of observations	1978	1978	1978	1978	1978	1978	

The data is from Prowess for listed companies during the period 2000–2019. The dependent variables are Delta cash, CAPEX, Other investments, Net share issues, Borrowings and Dividends. All the variables are defined in table 1. We include 2 lagged values of net cash flow from operations. We also add one-year lagged value of dependent variables. The regression equations are estimated with firm and year fixed effects. To save the space, year dummies are not reported. Robust standard errors are presented in parentheses. Coefficients significant at the 10%, 5%, and 1% levels are indicated by *, **, and ***, respectively. All equations are estimated using the single-equation model.

- 6. Drivers of internal cash flow allocations:** In the previous sections we have analysed how firms in general allocate their internal cash flow to competing uses. Our analysis reveals that cash flows are more sensitive to borrowings, but it also facilitates investments in subsequent years by borrowing. Our analysis also revealed that equity is not a major source of finance for the Indian firms. In the subsequent sections, we analyse some of the major determinants of the cash flow sensitivities. Our focus is on some of the firm level and country level factors that affect the sensitivity of internal cash flows to competing uses.

6.1 Financial flexibility versus Debt overhang: Our base line results indicate that firms use much of their internal cash flows for retiring the debt. Given that internal source of funds has competing uses, the high debt cash flow sensitivity can be looked from 2 lenses, 1. Financial flexibility, 2. Debt overhang. Financial flexibility is the ability of firms to invest in value-maximizing projects in a timely manner (Denis, 2011). Thus, firms may spare their borrowing power so that when an opportunity knocks, they can easily fund it by utilizing the untapped borrowings. Table-4 exhibits this pattern to a certain extent. Even though firms reduce their debt at the year "t", in subsequent years they borrow the funds for making investments. On the other hand, firms may be retiring their debt fearing the chances of debt overhang and such firms may continue to retire debt during subsequent years also. Such firms may be unable to increase investments in subsequent years. We try to analyse this formally by checking how debt levels (total debt divided by total assets) influence the allocation of cash flow to different uses.

Table 5 & 6 exhibit the results of cash flow allocations based on the levels of leverage. Basic difference between highly levered firms and less levered firm is that, former allocates Rs .616 ($0.567+0.038+0.011$) of current cash flows for making changes in capital structure and payment of dividend while latter uses Rs .518 ($0.196+0.127+0.195$) for increasing the assets base. However, for the less levered firms (more flexible firms), they allocate current cash flows for increasing their cash reserves (Rs .196) than for making CAPEX (Rs.127) unlike the more levered firms which allocate Rs .20 for CAPEX investments and only Rs 5 for increasing the cash reserves. This result is surprising since empirical evidence suggests that highly levered firms make less investments and increase their cash buffer (Denis & Sibilkov, 2010). Over a period of 3 years, total investments (CAPEX and other investments) made by more flexible firms is Rs .454 rupees which is less than the total investments made by relatively more levered firms (Rs .473). Flexible firms raise less funds from external markets via borrowings and draw down on their cash balances while highly levered firms use borrowings to increase their investments as their cash reserves are less. Highly levered firms pay Rs 2.5 as dividends for three years while flexible firms pay Rs 6.5 as dividends indicating that firms that are more levered prefer investments than paying dividends. With respect to controls, keeping other variables unchanged, an

increase in leverage in highly levered firms, leads to disinvestment (CAPEX and other investments becomes negative) while flexible firms, continues to make investments in CAPEX. Overall, the results related to leverage indicates that contrary to popular notion that high leverage deters firms from making investments, our results indicate that levered firms make more investments than flexible firms and much of this investments is channelled as CAPEX. As already mentioned, since bank debt is the major source of borrowing, it brings in monitoring efficiency which forces the firms to invest in real projects. Our results are consistent with recent research findings about the disciplining role of debt which enables firms with high debt overhang to invest relatively more than otherwise similar firms (Barbiero et al., 2018).

Table 5

Regression results based on leverage classification (More levered firms)

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Borrowings (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.054***	0.197***	0.044***	0.038***	0.567***	0.011***	0.911
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.03	0	
Net cash flow from operation at t+1	-0.018*	0.115***	0.014*	-0.009	- 0.117***	0.009***	-0.006
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.03	0	
Net cash flow from operation at t+2	-0.007	0.100***	0.003	-0.011	- 0.108***	0.005**	-0.018
Robust Standard Errors	-0.01	-0.02	-0.02	-0.01	-0.03	0	
MB ratio (3-Year Average)	0.001	0.028*	0.009**	- 0.016***	-0.024*	0.004***	0.002
Robust Standard Errors	0	-0.01	0	-0.01	-0.01	0	
Sales Growth (3-Year Average)	0.005	0.144***	0.007	- 0.034***	- 0.146***	0.010***	-0.014
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
Log (total assets) (3-Year Average)	- 0.015***	- 0.060***	-0.005*	0.013***	0.074***	- 0.002***	0.005
Robust Standard Errors	0	-0.01	0	0	-0.01	0	
Leverage (3-Year Average)	-0.019*	- 0.224***	-0.013	-0.025*	0.290***	- 0.017***	-0.008
Robust Standard Errors	-0.01	-0.04	-0.01	-0.01	-0.04	0	
Operating Profitability (3-Year Average)	0.005	0.052**	-0.017**	-0.004	-0.008	0.004***	0.032
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
Log(age) (3-Year Average)	0.018*	-0.079**	0.004	0.019*	0.031	0.001	-0.006
Robust Standard Errors	-0.01	-0.04	-0.01	-0.01	-0.04	0	
Constant	0.085***	0.851***	0.025	- 0.149***	- 0.838***	0.027***	0.001
Robust Standard Errors	-0.03	-0.12	-0.04	-0.04	-0.12	-0.01	
R-square	0.036	0.255	0.021	0.058	0.314	0.195	
Number of observations	6406	6406	6406	6406	6406	6406	

We split the overall sample into deciles based on the leverage (debt to total assets) in each year. We classify firms as highly levered firms when they fall in last four deciles based on their leverage. Firms in the first four deciles are classified as less levered firms. All the variables are defined in the table 1

Table 6

Regression results based on leverage classification (Less levered firms)

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Borrowings (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.196***	0.127***	0.195***	0.036***	0.356***	0.021***	0.931
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
Net cash flow from operation at t+1	- 0.043***	0.068***	0.005	0.005	-0.056***	0.026***	0.005
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
Net cash flow from operation at t+2	-0.013	0.063***	-0.004	-0.015**	-0.038*	0.018***	0.011
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
MB ratio (3-Year Average)	- 0.007***	0.008**	0.007**	-0.004	-0.014***	0.005***	-0.005
Robust Standard Errors	0	0	0	0	0	0	
Sales Growth (3-Year Average)	0.011	0.102***	-0.018*	- 0.039***	-0.063***	0.015***	0.008
Robust Standard Errors	-0.01	-0.01	-0.01	-0.01	-0.01	0	
Log (total assets) (3-Year Average)	-0.005	- 0.016***	-0.009**	0.009***	0.016***	-0.002	-0.005
Robust Standard Errors	0	-0.01	0	0	-0.01	0	
Leverage (3-Year Average)	-0.021	0.03	-0.038**	-0.029**	0.094***	-0.033***	0.003
Robust Standard Errors	-0.02	-0.03	-0.02	-0.01	-0.03	-0.01	
Operating Profitability (3-Year Average)	-0.007	0.024*	0.003	-0.008	-0.006	0.009***	0.015
Robust Standard Errors	-0.01	-0.01	-0.01	-0.01	-0.01	0	
Log(age) (3-Year Average)	0.003	-0.040**	-0.001	0.01	-0.011	0.018***	-0.021
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	-0.01	
constant	0.032	0.242***	0.044	- 0.099***	-0.102*	-0.02	0.097
Robust Standard Errors	-0.03	-0.06	-0.05	“(0.04)	-0.06	-0.02	
R-square	0.121	0.109	0.1	0.034	0.163	0.155	
Number of observations	6299	6299	6299	6299	6299	6299	

We split the overall sample into deciles based on the leverage (debt to total assets) in each year. We classify firms as low levered firms when they fall in last four deciles based on their leverage. Firms in the first four deciles are classified as less levered firms. All the variables are defined in the table 1

6.2 Large firms versus Small firms: Do large firms and small firms differ in their allocation of internal cash flows. While large firms are viewed as unconstrained firms in general, since they have better access to capital owing to less information asymmetry unlike the small firms (Acharya et al., 2007). Also, in the context of capital structure, (Frank & Goyal, 2003) show that large firms follow a pecking order for financing investments. Similarly (Rajan & Zingales, 1995) in their study reports the positive comovement between leverage and firm size. In the context of India, it is argued that, small firms have the more flexibility to make managerial decisions and lower response time to the business environment changes. However, larger firms have better economies of scales, garner better access to bank loans and has a strong political clout (De & Nagaraj, 2014). This could imply that larger the firms, more access to funds hence more investments unlike small firms which, due to capital constraints rely more on their saved cash to make the investments. Continuing with the same logic, the investment cash flow sensitivity of large firms would be relatively less than small firms. We analyse the cash flow sensitivities of the sample firms to different decisions classified based on firm size (Log of total assets) and table 7 & table 8 report our results. Both large firms and small firms, exhibit higher sensitivity of internal cash flows to the debt during the year " t ". The investment cash flow sensitivity of large firms is less than the small firms. Also, the allocation of cash flows during the subsequent years is different for large and small firms. Large firms increase their investments in subsequent years, and they raise the funds required for this via borrowings. Small firms raise less funds by the way of borrowings and as a result their investments in CAPEX is less than the large firms. Total investments made by large firms over a three period is Rs .53 and for small firms it is Rs .34. The total borrowings repaid by large firms over the 3 years period is Rs .273 and for small firms this is Rs .434 which means for small firm's major use of cash flow to make changes in financial structure. Large firms also maintain a stable dividend policy unlike small firms.

Table 7

Regression results based on firm size classification (Small firms)

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Borrowings (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.125***	0.162***	0.101***	0.046***	0.492***	0.013***	0.939
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.00)	
Net cash flow from operation at t+1	-0.040***	0.044**	0	0.018**	-0.029	0.014***	0.007
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Net cash flow from operation at t+2	-0.001	0.041**	-0.009	-0.006	-0.029	0.008***	0.004
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
MB ratio (3-Year Average)	-0.005*	0.018**	0.008***	-0.004	-0.029***	0.006***	-0.006
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Sales Growth (3-Year Average)	0.023***	0.108***	0.005	-0.033***	-0.110***	0.008***	0.001
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Log (total assets) (3-Year Average)	-0.010***	-0.025***	-0.001	0.003	0.036***	(0.00)	0.003
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Leverage (3-Year Average)	-0.023**	-0.027	-0.021**	-0.018**	0.124***	-0.031***	0.004
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Operating Profitability (3-Year Average)	0.001	0.030*	-0.001	-0.015**	0.007	0.006***	0.028
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Log(age) (3-Year Average)	-0.006	-0.076***	-0.009	0.021	0.049	0.005	-0.07
Robust Standard Errors	(0.01)	(0.03)	(0.01)	(0.01)	(0.03)	(0.00)	
Constant	0.077**	0.380***	0.01	-0.074*	-0.359***	-0.005	0.029
Robust Standard Errors	(0.04)	(0.09)	(0.04)	(0.04)	(0.01)	(0.01)	
R-square	0.091	0.105	0.068	0.037	0.264	0.121	
Number of observations	5820	5820	5820	5820	5820	5820	

We split the overall sample into deciles based on the size (log of total assets) in each year. We classify firms as small firms when they fall in first four deciles based on their size. Firms in the bottom four deciles are classified as large firms. All the variables are defined in the table 1

Table 8

Regression results based on firm size classification (Large firms)

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Borrowings (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.085***	0.145***	0.129***	0.027***	0.513***	0.019***	0.918
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.03	0	
Net cash flow from operation at t+1	0	0.104***	0.021*	-0.008	-0.118***	0.018***	0.017
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.03	0	
Net cash flow from operation at t+2	-0.025*	0.119***	0.008	-0.016*	-0.121***	0.017***	-0.018
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.03	0	
MB ratio (3-Year Average)	-0.001	0.017***	0.005*	-0.009***	-0.019***	0.006***	-0.001
Robust Standard Errors	0	-0.01	0	0	-0.01	0	
Sales Growth (3-Year Average)	0.011	0.127***	0.008	-0.045***	-0.118***	0.013***	-0.004
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
Log (total assets) (3-Year Average)	-0.013***	-0.037***	-0.004	0.017***	0.039***	-0.002*	-0.024
Robust Standard Errors	0	-0.01	0	0	-0.01	0	
Leverage (3-Year Average)	-0.01	-0.144***	-0.017	-0.015	0.198***	-0.026***	-0.014
Robust Standard Errors	-0.01	-0.03	-0.01	-0.01	-0.03	0	
Operating Profitability (3-Year Average)	-0.002	0.058***	-0.014	0.008	-0.040*	0.007**	0.017
Robust Standard Errors	-0.01	-0.02	-0.01	-0.01	-0.02	0	
Log(age) (3-Year Average)	0.009	-0.069***	0.01	0.015*	0.028	0.009**	0.002
Robust Standard Errors	-0.01	-0.03	-0.01	-0.01	-0.03	0	
Constant	0.099***	0.632***	0.004	-0.209***	-0.534***	0.012	0.004
Robust Standard Errors	-0.03	-0.1	-0.04	-0.04	-0.11	-0.01	
R-square	0.042	0.265	0.058	0.068	0.252	0.214	
Number of observations	6793	6793	6793	6793	6793	6793	

We split the overall sample into deciles based on the size (log of total assets) in each year. We classify firms as small firms when they fall in first four deciles based on their size. Firms in the bottom four deciles are classified as large firms. All the variables are defined in the table 1

6.3 Group affiliated firms versus Stand-alone firms: Table 8 & 9 exhibits the cash flow sensitivities of group affiliated firms and stand-alone firms. As explained in section 2.3, empirical findings support the evidence that group affiliated firms are less constrained than stand-alone firms since they have better access to capital markets (Lensink et al., 2003). In our analysis, both stand alone firms and group affiliated firms allocate more than half of their current cash flows for the repayment of borrowings. The investments cash flow sensitivity of group affiliated firms is slightly lower than the stand-alone firms similarly cash flow the sensitivity of cash which is relatively lower for the group affiliated firms. Both these sensitivities imply group affiliated firms are less constrained than stand-alone firms based on the studies (Ağca & Mozumdar, 2008) and (Almeida et al., 2004). Group affiliated firms have better access to borrowings, and this is reflected in the investments of affiliated firms which makes Rs 21 in CAPEX by borrowing Rs 20. Standalone firms invest Rs 12 in CAPEX and they borrow Rs 9 for this and draw down their cash reserves. However, when the leverage increases, group affiliated firms disinvest more than standalone firms and use the proceeds to reduce the debt burden.

Table 8

Group affiliated firms

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Borrowings (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.087***	0.147***	0.134***	0.030***	0.508***	0.020***	0.926
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.00)	
Net cash flow from operation at t+1	0.005	0.092***	0.012	0.004	-0.124***	0.018***	0.007
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Net cash flow from operation at t+2	-0.034***	0.120***	-0.003	-0.024***	-0.087***	0.013***	-0.015
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.00)	
MB ratio (3-Year Average)	-0.001	0.023***	0.006*	-0.008**	-0.022***	0.006***	0.004
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Sales Growth (3-Year Average)	0.001	0.108***	0.001	-0.032***	-0.099***	0.014***	-0.007
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Log (total assets) (3-Year Average)	-0.010***	-0.038***	-0.009**	0.016***	0.042***	-0.002**	-0.001
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Leverage (3-Year Average)	-0.019**	-0.079***	-0.021**	-0.029***	0.167***	-0.029***	-0.0107
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Operating Profitability (3-Year Average)	0.002	0.066***	-0.006	0.005	-0.054**	0.008***	0.021
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Log(age) (3-Year Average)	0	-0.045*	0.011	0.015	0.006	0.006*	-0.007
Robust Standard Errors	(0.01)	(0.03)	(0.01)	(0.01)	(0.03)	(0.00)	
Constant	0.091***	0.503***	0.034	-0.176***	-0.434***	0.013	0.031
Robust Standard Errors	(0.02)	(0.1)	(0.04)	(0.04)	(0.09)	(0.01)	
R-square	0.049	0.213	0.063	0.054	0.243	0.215	
Number of observations	6752	6752	6752	6752	6752	6752	

We split the overall sample based on the ownership category. We classify firms as group affiliated ones if they belong to business groups and standalone otherwise.

Table 8

Standalone firms

	Delta Cash (1)	CAPEX (2)	Other Investments (3)	Net share issues (4)	Borrowings (5)	Dividends (6)	Sum (7)
Net cash flow from operation	0.123***	0.173***	0.082***	0.043***	0.495***	0.015***	0.931
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Net cash flow from operation at t+1	-0.031***	0.078***	-0.002	0.003	-0.054***	0.016***	0.01
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
Net cash flow from operation at t+2	0.002	0.055***	-0.004	-0.009	-0.049**	0.010***	0.005
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.00)	
MB ratio (3-Year Average)	-0.001	0.016**	0.006**	-0.007***	-0.025***	0.005***	-0.006
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Sales Growth (3-Year Average)	0.020***	0.145***	0.005	-0.043***	-0.142***	0.010***	-0.005
Robust Standard Errors	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.00)	
Log (total assets) (3-Year Average)	-0.012***	-0.037***	0	0.010***	0.044***	-0.002**	0.003
Robust Standard Errors	(0.00)	(0.01)	(0.00)	(0.00)	(0.01)	(0.00)	
Leverage (3-Year Average)	-0.014	-0.049**	-0.015	-0.008	0.119***	-0.025***	0.008
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.00)	
Operating Profitability (3-Year Average)	-0.006	0.035***	-0.005	-0.015**	-0.003	0.007***	0.019
Robust Standard Errors	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	0	
Log(age) (3-Year Average)	0.003	-0.067***	0	0.012	0.034	0.007**	-0.011
Robust Standard Errors	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.00)	
Constant	0.071**	0.480***	-0.01	-0.110***	-0.428***	0.008	0.011
Robust Standard Errors	(0.03)	(0.07)	(0.03)	(0.03)	(0.08)	(0.01)	
R-square	0.082	0.165	0.041	0.048	0.26	0.149	
Number of observations	8418	8418	8418	8418	8418	8418	

We split the overall sample based on the ownership category. We classify firms as group affiliated ones if they belong to business groups and standalone otherwise

7. **Conclusion:** Our study analyses how firms in an emerging country, India, allocates their internal cash flows to various uses. Owing to capital market imperfections, internal cash flows are vital for firms in credit constrained firms. Cash flows are overly sensitive to borrowings & the major use of internal cash flow is for repayment of borrowings. Faced with a positive cash flow, firms allocate more than 50% of funds for the repayment of debt. Regardless of the firms-characteristics in terms of size, leverage and group-affiliation, firms on an average use contemporaneous cash flow to repay the borrowings. However, in the subsequent periods, firms use borrowings to increase both their CAPEX and other investments. Our study provides evidence that borrowings are major source of funds for facilitating investments unlike equity issues which is not a major source of finance. Firms that are unconstrained in terms of size and access to internal capital markets can borrow more and increase the investments. However, our study also finds that financially flexible firms invests less as compared to firms that are highly levered. Overall our study finds that borrowings is a major source of finance in countries like India where private debt in the form of borrowings from the bank is cheaper than the cost of raising funds via equity issues apart from the tax-shield benefits of the borrowings. Since, large sized, group affiliated firms wield a relative advantage in debt market, such firms are able to invest more.

References

- Acharya, V. V., Almeida, H., & Campello, M. (2007). Is cash negative debt? A hedging perspective on corporate financial policies. *Journal of Financial Intermediation*, 16(4), 515–554. <https://doi.org/10.1016/j.jfi.2007.04.001>
- Ağca, Ş., & Mozumdar, A. (2008). The impact of capital market imperfections on investment–cash flow sensitivity. *Journal of Banking & Finance*, 32(2), 207–216. <https://doi.org/10.1016/j.jbankfin.2007.02.013>
- Allen, F., Chakrabarti, R., De, S., Qian, J. “QJ,” & Qian, M. (2012). Financing firms in India. *Journal of Financial Intermediation*, 21(3), 409–445. <https://doi.org/10.1016/j.jfi.2012.01.003>
- Almeida, H., Campello, M., & Weisbach, M. S. (2004). The Cash Flow Sensitivity of Cash. *The Journal of Finance*, 59(4), 1777–1804. <https://doi.org/10.1111/j.1540-6261.2004.00679.x>
- Almeida, H., Campello, M., & Weisbach, M. S. (2011). Corporate financial and investment policies when future financing is not frictionless. *Journal of Corporate Finance*, 17(3), 675–693. <https://doi.org/10.1016/j.jcorpfin.2009.04.001>
- Athey, M. J., & Laumas, P. S. (1994). Internal funds and corporate investment in India. *Journal of Development Economics*, 45(2), 287–303. [https://doi.org/10.1016/0304-3878\(94\)90034-5](https://doi.org/10.1016/0304-3878(94)90034-5)
- Baghai, R. P., & Becker, B. (2018). Non-rating revenue and conflicts of interest. *Journal of Financial Economics*, 127(1), 94–112. <https://doi.org/10.1016/j.jfineco.2017.10.004>
- Banerjee, A. V., & Duflo, E. (2014). Do Firms Want to Borrow More? Testing Credit Constraints Using a Directed Lending Program. *The Review of Economic Studies*, 81(2), 572–607. <https://doi.org/10.1093/restud/rdt046>

- Bao, D., Chan, K. C., & Zhang, W. (2012). Asymmetric cash flow sensitivity of cash holdings. *Journal of Corporate Finance*, 18(4), 690–700.
<https://doi.org/10.1016/j.jcorpfin.2012.05.003>
- Barbiero, F., Popov, A. A., & Wolski, M. (2018). *Debt Overhang and Investment Efficiency* (SSRN Scholarly Paper ID 3138846). Social Science Research Network.
<https://papers.ssrn.com/abstract=3138846>
- Bertrand, M., Mehta, P., & Mullainathan, S. (2002). FERRETING OUT TUNNELING: AN APPLICATION TO INDIAN BUSINESS GROUPS. *QUARTERLY JOURNAL OF ECONOMICS*, 28.
- Bhattacharya, S. (1979). Imperfect Information, Dividend Policy, and “The Bird in the Hand” Fallacy. *The Bell Journal of Economics*, 10(1), 259. <https://doi.org/10.2307/3003330>
- Brockman, P., & Chung, D. Y. (2001). Managerial timing and corporate liquidity: Evidence from actual share repurchases. *Journal of Financial Economics*, 61(3), 417–448.
[https://doi.org/10.1016/S0304-405X\(01\)00068-X](https://doi.org/10.1016/S0304-405X(01)00068-X)
- Brunnermeier, M. K., Eisenbach, T. M., & Sannikov, Y. (2011). *Macroeconomics with Financial Frictions: A Survey*. 92.
- Bushman, R. M., Smith, A. J., & Zhang, F. (2011). Investment Cash Flow Sensitivities Really Reflect Related Investment Decisions. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.842085>
- Byoun, S. (2007). *Financial Flexibility, Leverage, and Firm Size*. 38.
- Byoun, S. (2008). How and When Do Firms Adjust Their Capital Structures toward Targets? *The Journal of Finance*, 63(6), 3069–3096. <https://doi.org/10.1111/j.1540-6261.2008.01421.x>
- Cao, D., Lorenzoni, G., & Walentin, K. (2013). *Financial Frictions, Investment and Tobin's*
- . 34.

- Chang, X., Dasgupta, S., Wong, G., & Yao, J. (2014). Cash-Flow Sensitivities and the Allocation of Internal Cash Flow. *Review of Financial Studies*, 27(12), 3628–3657.
<https://doi.org/10.1093/rfs/hhu066>
- Chauhan, G. S. (2017). Corporate financing and deleveraging of firms in India. *IIMB Management Review*, 29(3), 170–187. <https://doi.org/10.1016/j.iimb.2017.06.002>
- Chauhan, Y., Pathak, R., & Kumar, S. (2018). Do bank-appointed directors affect corporate cash holding? *International Review of Economics & Finance*, 53, 39–56.
<https://doi.org/10.1016/j.iref.2017.10.009>
- Chen, H. (Jason), & Chen, S. (Jenny). (2012). Investment-cash flow sensitivity cannot be a good measure of financial constraints: Evidence from the time series. *Journal of Financial Economics*, 103(2), 393–410. <https://doi.org/10.1016/j.jfineco.2011.08.009>
- Cleary, S., Povel, P., & Raith, M. (2007). The U-Shaped Investment Curve: Theory and Evidence. *The Journal of Financial and Quantitative Analysis*, 42(1), 1–39. JSTOR.
<https://www.jstor.org/stable/27647284>
- Daniel, N. D., Denis, D. J., & Naveen, L. (2008). *Sources of Financial Flexibility: Evidence from Cash Flow Shortfalls*. 52.
- Das, S., SDas@imf.org, Tulin, V., & VTulin@imf.org. (2017). Financial Frictions, Underinvestment, and Investment Composition: Evidence from Indian Corporates. *IMF Working Papers*, 17(134), 1. <https://doi.org/10.5089/9781484302644.001>
- Dasgupta, S., Noe, T. H., & Wang, Z. (2011). Where Did All the Dollars Go? The Effect of Cash Flows on Capital and Asset Structure. *The Journal of Financial and Quantitative Analysis*, 46(5), 1259–1294. JSTOR.
<https://www.jstor.org/stable/41409650>
- De, P. K., & Nagaraj, P. (2014). Productivity and firm size in India. *Small Business Economics*, 42(4), 891–907. <https://doi.org/10.1007/s11187-013-9504-x>

- DeAngelo, H., & DeAngelo, L. (2006). The irrelevance of the MM dividend irrelevance theorem. *Journal of Financial Economics*, 79(2), 293–315.
<https://doi.org/10.1016/j.jfineco.2005.03.003>
- Denis, D. J. (2011). Financial flexibility and corporate liquidity. *Journal of Corporate Finance*, 17(3), 667–674. <https://doi.org/10.1016/j.jcorpfin.2011.03.006>
- Denis, D. J., & Sibilkov, V. (2010). Financial Constraints, Investment, and the Value of Cash Holdings. *Review of Financial Studies*, 23(1), 247–269.
<https://doi.org/10.1093/rfs/hhp031>
- Desai, M. A., Foley, C. F., & Hines, J. R. (2004). A Multinational Perspective on Capital Structure Choice and Internal Capital Markets. *The Journal of Finance*, 59(6), 2451–2487. <https://doi.org/10.1111/j.1540-6261.2004.00706.x>
- Drobetz, W., Haller, R., Meier, I., & Tarhan, V. (2017). The impact of liquidity crises on cash flow sensitivities. *The Quarterly Review of Economics and Finance*, 66, 225–239.
<https://doi.org/10.1016/j.qref.2017.03.004>
- Drobetz, W., Janzen, M., & Meier, I. (2019). Investment and financing decisions of private and public firms. *Journal of Business Finance & Accounting*, 46(1–2), 225–262.
<https://doi.org/10.1111/jbfa.12367>
- Easterbrook, F. H. (1984). *Two Agency-Cost Explanations of Dividends*. 11.
- Faccio, M., Marchica, M.-T., & Mura, R. (2016). CEO gender, corporate risk-taking, and the efficiency of capital allocation. *Journal of Corporate Finance*, 39, 193–209.
<https://doi.org/10.1016/j.jcorpfin.2016.02.008>
- Farre-Mensa, J., Michaely, R., & Schmalz, M. C. (2014). Financing Payouts. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2535675>

- Faulkender, M., Flannery, M. J., Hankins, K. W., & Smith, J. M. (2012). Cash flows and leverage adjustments. *Journal of Financial Economics*, 103(3), 632–646.
<https://doi.org/10.1016/j.jfineco.2011.10.013>
- Fazzari, S. M., Hubbard, R. G., Petersen, B. C., Blinder, A. S., & Poterba, J. M. (1988). Financing Constraints and Corporate Investment. *Brookings Papers on Economic Activity*, 1988(1), 141–206. JSTOR. <https://doi.org/10.2307/2534426>
- Frank, M. Z., & Goyal, V. K. (2003). Testing the pecking order theory of capital structure\$. *Journal of Financial Economics*, 32.
- Gatchev, V. A., Pulvino, T., & Tarhan, V. (2010). The Interdependent and Intertemporal Nature of Financial Decisions: An Application to Cash Flow Sensitivities. *The Journal of Finance*, 65(2), 725–763. <https://doi.org/10.1111/j.1540-6261.2009.01549.x>
- Gatchev, V. A., Spindt, P. A., & Tarhan, V. (2009a). How do firms finance their investments? *Journal of Corporate Finance*, 15(2), 179–195.
<https://doi.org/10.1016/j.jcorpfin.2008.11.001>
- Gatchev, V. A., Spindt, P. A., & Tarhan, V. (2009b). How do firms finance their investments?: The relative importance of equity issuance and debt contracting costs. *Journal of Corporate Finance*, 15(2), 179–195.
<https://doi.org/10.1016/j.jcorpfin.2008.11.001>
- Gopalan, R., Gopalan, Y., & Koharki, K. (2019). *Assessing the Quality of Bank Loan Ratings*. 60.
- Gopalan, R., Nanda, V., & Seru, A. (2007). Affiliated firms and financial support: Evidence from Indian business groups. *Journal of Financial Economics*, 86(3), 759–795.
<https://doi.org/10.1016/j.jfineco.2006.09.008>

- Grullon, G., & Michaely, R. (2002). Dividends, Share Repurchases, and the Substitution Hypothesis. *The Journal of Finance*, 57(4), 1649–1684. <https://doi.org/10.1111/1540-6261.00474>
- Harford, J., Klasa, S., & Maxwell, W. F. (2014). Refinancing Risk and Cash Holdings: Refinancing Risk and Cash Holdings. *The Journal of Finance*, 69(3), 975–1012. <https://doi.org/10.1111/jofi.12133>
- Hennessy, C. A., & Whited, T. M. (2007). How Costly Is External Financing? Evidence from a Structural Estimation. *The Journal of Finance*, 62(4), 1705–1745. <https://doi.org/10.1111/j.1540-6261.2007.01255.x>
- Hovakimian, A., Opler, T., & Titman, S. (2001). The Debt-Equity Choice. *The Journal of Financial and Quantitative Analysis*, 36(1), 1–24. JSTOR. <https://doi.org/10.2307/2676195>
- Huang, R., & Ritter, J. R. (2004). *Testing the Market Timing Theory of Capital Structure*. 56.
- Huang, R., & Ritter, J. R. (2015). Corporate Cash Shortfalls and Financing Decisions. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2589096>
- Jensen, M. C. (1976). *Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure*. 78.
- Jensen, M. C. (1986). Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers. *The American Economic Review*, 76(2), 323–329. <https://www.jstor.org/stable/1818789>
- Kali, R., & Sarkar, J. (2011). Diversification and tunneling: Evidence from Indian business groups. *Journal of Comparative Economics*, 39(3), 349–367. <https://doi.org/10.1016/j.jce.2011.04.001>

- Kaplan, S. N., & Zingales, L. (1997). Do Investment-Cash Flow Sensitivities Provide Useful Measures of Financing Constraints? *The Quarterly Journal of Economics*, 112(1), 169–215. <https://doi.org/10.1162/003355397555163>
- Keynes, J. M. (1936). *The General Theory of Employment, Interest, and Money*. 263.
- Khanna, T., & Palepu, K. (2000). Is Group Affiliation Profitable in Emerging Markets? An Analysis of Diversified Indian Business Groups. *The Journal of Finance*, 55(2), 867–891. <https://doi.org/10.1111/0022-1082.00229>
- Khanna, V. S., & Varottil, U. (2012). Developing the Market for Corporate Bonds in India. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2021602>
- Khurana, I. K., Martin, X., & Pereira, R. (2006). Financial Development and the Cash Flow Sensitivity of Cash. *The Journal of Financial and Quantitative Analysis*, 41(4), 787–807. JSTOR. <https://www.jstor.org/stable/27647274>
- Koijen, R. S. J., & Van Nieuwerburgh, S. (2011). Predictability of Returns and Cash Flows. *Annual Review of Financial Economics*, 3(1), 467–491. <https://doi.org/10.1146/annurev-financial-102710-144905>
- Komera, S., Jijo Lukose, P. J., & Sasidharan, S. (2018). Does business group affiliation encourage R&D activities? Evidence from India. *Asia Pacific Journal of Management*, 35(4), 887–917. <https://doi.org/10.1007/s10490-017-9530-3>
- Lambrecht, B. M., & Myers, S. C. (2013). *The Dynamics of Investment, Payout and Debt*. 50.
- Lang, L. H. P., Stulz, RenéM., & Walkling, R. A. (1989). Managerial performance, Tobin's Q, and the gains from successful tender offers. *Journal of Financial Economics*, 24(1), 137–154. [https://doi.org/10.1016/0304-405X\(89\)90075-5](https://doi.org/10.1016/0304-405X(89)90075-5)
- Larkin, Y., Ng, L., & Zhu, J. (2018). The fading of investment-cash flow sensitivity and global development. *Journal of Corporate Finance*, 50, 294–322. <https://doi.org/10.1016/j.jcorpfin.2018.04.003>

- Lensink, R., van der Molen, R., & Gangopadhyay, S. (2003). Business groups, financing constraints and investment: The case of India. *Journal of Development Studies*, 40(2), 93–119. <https://doi.org/10.1080/00220380412331293787>
- Lewellen, J., & Lewellen, K. (2016). Investment and Cash Flow: New Evidence. *Journal of Financial and Quantitative Analysis*, 51(4), 1135–1164. <https://doi.org/10.1017/S002210901600065X>
- Miller, M. H., & Modigliani, F. (1961). Dividend Policy, Growth, and the Valuation of Shares. *The Journal of Business*, 34(4), 411–433. <https://www.jstor.org/stable/2351143>
- Modigliani, F., & Miller, M. H. (1958). The Cost of Capital, Corporation Finance and the Theory of Investment. *The American Economic Review*, 48(3), 261–297. <http://www.jstor.org/stable/1809766>
- Moshirian, F., Nanda, V., Vadilyev, A., & Zhang, B. (2017). What drives investment–cash flow sensitivity around the World? An asset tangibility Perspective. *Journal of Banking & Finance*, 77, 1–17. <https://doi.org/10.1016/j.jbankfin.2016.12.012>
- Myers, S. C. (1977). Determinants of corporate borrowing. *Journal of Financial Economics*, 5(2), 147–175. [https://doi.org/10.1016/0304-405X\(77\)90015-0](https://doi.org/10.1016/0304-405X(77)90015-0)
- Myers, S. C., & Majluf, N. S. (1984). *Corporate Financing and Investment Decisions when Firms Have Information that Investors Do Not Have*.
- Pulvino, T., & Tarhan, V. (2006). *Cash Flow Sensitivities With Constraints*. 45.
- Rajan, R. G., & Zingales, L. (1995). What Do We Know about Capital Structure? Some Evidence from International Data. *The Journal of Finance*, 50(5), 1421–1460. <https://doi.org/10.1111/j.1540-6261.1995.tb05184.x>
- Riddick, L. A., & Whited, T. M. (2009). The Corporate Propensity to Save. *The Journal of Finance*, 64(4), 1729–1766. <https://doi.org/10.1111/j.1540-6261.2009.01478.x>

- Servaes, H. (1991). Tobin's Q and the Gains from Takeovers. *The Journal of Finance*, 46(1), 409–419. JSTOR. <https://doi.org/10.2307/2328702>
- Srinivasan, A., & Thampy, A. (2017). The effect of relationships with government-owned banks on cash flow constraints: Evidence from India. *Journal of Corporate Finance*, 46, 361–373. <https://doi.org/10.1016/j.jcorpfin.2017.07.007>
- Vig, V. (2013). Access to Collateral and Corporate Debt Structure: Evidence from a Natural Experiment: Access to Collateral and Corporate Debt Structure. *The Journal of Finance*, 68(3), 881–928. <https://doi.org/10.1111/jofi.12020>
- Whited, T. M. (1992). Debt, Liquidity Constraints, and Corporate Investment: Evidence from Panel Data. *The Journal of Finance*, 47(4), 1425–1460. JSTOR. <https://doi.org/10.2307/2328946>